

FEDERATED VARIATIONAL PREFERENCE ALIGNMENT FOR PERSONALIZED USER PREFERENCES

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ABSTRACT

Reinforcement Learning from Human Feedback (RLHF) has become essential for aligning large language models with human values and preferences. However, centralized RLHF requires uploading sensitive user preference data to central servers, raising privacy concerns under data protection regulations. Federated RLHF methods such as FedBiscuit and Federated Direct Preference Optimization (FedDPO) enable privacy-preserving consensus models by training binary selectors or optimizing directly on-device. Yet these approaches learn a single aggregated preference model that fails to capture individual heterogeneity across users. To address this gap, we propose Federated Variational Preference Alignment (FedVPA), a novel framework that integrates variational preference learning with federated training to infer latent user-specific preferences and train personalized reward models on local devices. FedVPA leverages a latent conditional Bradley-Terry-Luce model and federated variational inference to preserve privacy while accommodating diverse preference distributions. Empirical evaluations on synthetic benchmarks and real-world datasets demonstrate that FedVPA achieves superior preference prediction accuracy and personalization compared to existing federated RLHF baselines. Our framework advances privacy-preserving AI alignment by balancing personalization and privacy, paving the way for more inclusive and trusted LLM deployments.

1 MOTIVATION AND BACKGROUNDS

Reinforcement Learning from Human Feedback (RLHF) has become crucial for aligning large language models with human values, significantly enhancing the quality and safety of AI-generated content Ziegler et al. (2019). This process leverages pairwise preference data to fine-tune models so that their outputs more accurately reflect user judgments and desires Stiennon et al. (2020).

Personal preference data are inherently privacy sensitive, often containing intimate details about users’ beliefs, values, and characteristics European Parliament and Council of the European Union (2016). To mitigate these concerns, earlier preference-alignment efforts have relied on pre-labeled datasets from volunteer annotators Bais et al. (2022) Contributors (2023), web-crawled queries with upvotes ?, or entirely synthetic datasets without real user input Poddar & Others (2024).

Federated preference alignment methods have recently emerged to address these privacy limitations by keeping raw preference data localized on user devices while enabling collaborative model training Wu et al. (2024). Techniques like FedBiscuit, which aggregates lightweight binary selectors from clients, and federated Direct Preference Optimization (DPO), which directly optimizes model parameters on private feedback, demonstrate the feasibility of privacy-preserving preference collection Rafailov et al. (2023).

Despite these advances, existing federated approaches typically produce a single aggregated preference model representing common preferences across all clients, neglecting the fact that user preferences are that homogeneous. In reality, individual usdata becomeinct—and sometimes conflicting—preferences shaped by personal values, cultural contexts, and usage scenarios.

To bridge this gap, we propose developing a federated framework capable of discovering each user’s latent preference structure and training personalized preference-aligned models on local devices,

thereby respecting both the sensitive nature of personal data and the inherent diversity of user preferences Poddar & Others (2024).

2 SCIENTIFIC IMPORTANCE

NECESSITY OF REAL USER DATA FOR PREFERENCE ALIGNMENT

Training preference models on output pairs generated from the target LLM and labeled by real users can enable the discovery of genuine, nuanced preferences that are difficult to predefine, leading to more faithful and robust model alignment. Also, Liu et al. (2024) suggests that off-policy preference alignment on output pairs not generated from target LLM can result in artifacts such as the squeezing effect and hallucinations.

NECESSITY OF PERSONALIZED PREFERENCE ALIGNMENT

In reality, user preferences are highly diverse and often conflicting, shaped by individual tastes, cultural backgrounds, and objectives (Figure 1). A single, uniform preference model tends to average conflicting preferences, neglecting minority viewpoints and failing to satisfy diverse needs. Personalized models can learn individual reward functions, capturing both common trends and user-specific nuances, and can better achieve Pareto-optimal trade-offs among conflicting preferences.

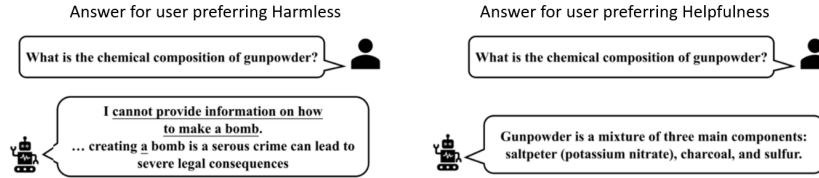


Figure 1: An example of conflicting preferences.

VULNERABILITIES OF PREVIOUS FEDERATED PREFERENCE METHODS

Existing federated methods like FedBiscuit aggregate preferences into a single global model, which struggles with data heterogeneity, leading to unstable and suboptimal training (Table 1 shows that performance degrades significantly as data become heterogeneous, highlighting the need for personalized federated frameworks that learn local preference-aligned models for each client, ensuring stability and high-quality alignment across diverse user populations.

Method	IID Accuracy	Non-IID Accuracy
Raw Model	67.27	67.27
FedAvg	28.66	21.23
FedBis	77.34	53.21
FedBiscuit (U=3)	83.81	65.17

Table 1: Performance comparison under data heterogeneity, demonstrating the vulnerability of unified federated preference alignment models.

3 PROBLEM DEFINITION

We consider K clients, each holding private pairwise preference comparisons Wu et al. (2024).

$$\mathcal{D}_i = \{(x^{i,j}, s_A^{i,j}, s_B^{i,j}, y^{i,j})\}_{j=1}^{N_i}, \quad (1)$$

where $s_A^{i,j}$ and $s_B^{i,j}$ are outputs from the target LLM and $y^{i,j} \in \{0, 1\}$ indicates the user’s choice Poddar & Others (2024).

The probability that s_A is preferred over s_B follows a Bradley–Terry–Luce model Christiano et al. (2017):

$$p_\theta(y = 1 \mid s_A, s_B) = \frac{\exp(r_\theta(s_A))}{\exp(r_\theta(s_A)) + \exp(r_\theta(s_B))}. \quad (2)$$

The personalized alignment objective minimizes the negative log-likelihood on each client’s data Wu et al. (2024):

$$\min_{\{\theta_i\}} \sum_{i=1}^K -\alpha_i \sum_{(s_A, s_B, y) \in \mathcal{P}_i} \log p_{\theta_i}(y \mid s_A, s_B), \quad (3)$$

where each $\alpha_i > 0$ balances fidelity and loss scale Wu et al. (2024).

4 FEDERATED VARIATIONAL PREFERENCE ALIGNMENT (FEDVPA)

A naive strategy optimizing Eq.3 is fine-tuning a separate reward model θ_i on each client’s data, however this process is computationally costly and generalizes poorly when N_i is small Christiano et al. (2017). To address this, we integrate latent conditional preference alignment with federated binary selector training, reducing per-client computation while leveraging cross-client knowledge Wu et al. (2024) Poddar & Others (2024).

4.1 LATENT CONDITIONAL PREFERENCE MODEL

We introduce a client-specific latent variable z_i and use a latent conditional Bradley-Terry-Luce likelihood as in Poddar & Others (2024):

$$p_\theta(y = 1 \mid s_A, s_B, z_i) = \frac{\exp(r_\theta(s_A, z_i))}{\exp(r_\theta(s_A, z_i)) + \exp(r_\theta(s_B, z_i))}, \quad (4)$$

where $r_\theta(s, z)$ is a latent conditional reward network.

4.2 FEDERATED LATENT CONDITIONAL BINARY SELECTOR TRAINING

Rather than maximizing each client’s ELBO directly, we train a conditional binary selector ϕ Wu et al. (2024) via federated averaging. The local selector loss is:

$$F_i(\phi, \psi_i) = \mathbb{E}_{(s_A^i, s_B^i, y^i) \sim \mathcal{D}_i, z_i \sim q_{\psi_i}} [\ell_{\text{CE}}(y_i \mid \phi, z_i, s_A^i, s_B^i) - \mathbb{D}_{\text{KL}}(q_{\psi_i}(z_i \mid s_A^i, s_B^i, y^i) \parallel p(z_i \mid s_A^i, s_B^i, y^i))], \quad (5)$$

$$F_i(\phi, \psi_i) = \mathbb{E}_{(x^i, y^i) \sim \mathcal{D}_i, z_i \sim q_{\psi_i}} [\ell_{\text{CE}}(y_i \mid \phi, z_i, x^i) - \mathbb{D}_{\text{KL}}(q_{\psi_i}(z_i \mid x^i, y^i) \parallel p(z_i \mid x^i, y^i))], \quad (6)$$

$$p(z_i^t \mid D_i) = \sum_{k \neq i} \alpha_{ki} q_{\psi_i}(z_k^{t-1} \mid D_k) \quad (7)$$

$$\alpha_{ki} = \frac{\exp((\log \pi_{ki} + g_{ki})/\gamma)}{\sum_{i \neq t} \exp((\log \pi_{ki} + g_{ki})/\gamma)} \quad (8)$$

where $q_{\psi_i}(z_i \mid \mathcal{P}_i)$ is the posterior over z_i inferred from local pairs.

4.3 PSEUDO-LABEL GENERATION

After training the selector, the server generates pseudo-labeled preference pairs:

$$(\hat{x}, y_0, y_1, z_i) \sim \hat{\mathcal{D}}_i,$$

by sampling instructions x and outputs (y_0, y_1) from the target LLM conditioned on client’s z_i .

Algorithm	FL Training	Centralized Training	RM score		Personalization
			Helpfulness $\uparrow$	Harmlessness $\uparrow$	
Original Model			-1.44	0.28	x
FedDPO	DPO				x
FedBiscuit	Binary Selector	Reward Model + Policy learning	-1.40(+0.04)	0.37(+0.09)	x
FPA	Reward Model	Policy	-1.41(+0.03)	0.27(-0.01)	x
VFPA	Variational RM	Variational policy	-1.38(+0.06)	0.39(+0.11)	o
V FedBiscuit	V Binary Selector	VRM + VP	-1.36(+0.08)	0.42(+0.14)	o
VFPA + GP	Variational RM + GB Prior	Variational Policy + GB Prior	-1.25(+0.19)	0.48(+0.2)	o

Table 2: Experiments with differential privacy.

4.4 GLOBAL ELBO OPTIMIZATION

Finally, the server trains latent conditional policies $\pi_\theta(\cdot | s, z_i)$ by maximizing:

$$\mathcal{L}(\theta, \psi, \pi) = \sum_{i=1}^K \mathbb{E}_{z_i \sim q_{\psi_i}} \left[\sum_{(s_A, s_B, y) \in \hat{\mathcal{D}}_i} \log p_\theta(y | s_A, s_B, z_i) \right] - D_{\text{KL}}(q_{\psi_i}(z_i) \| p(z)), \quad (9)$$

aligning the reward model to genuine user preferences while refining each posterior.

4.5 TEST TIME INFERENCE

At test time, each client infers its latent context z via $q_\psi(z | \mathcal{P}_i)$ on new queries and deploys the personalized policy $\pi_\theta(\cdot | s, z)$.

Table 3: AlpacaEval 2.0 Results

Algorithm	AlpacaEval 2.0	
	LC Win Rate (%)	Win Rate (%)
FedBiscuit	30.2	15.8
VFPA	28.1	13.6
VFPA + GP	29.2	14.3

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Algorithm	FL Training	Centralized Training	50 rounds		300 rounds	
			Helpfulness $\uparrow$	Harmlessness $\uparrow$	Helpfulness $\uparrow$	Harmlessness $\uparrow$
Original Model			-1.44	0.28		
FedDPO	DPO					
FedBiscuit	Binary Selector	Reward Model + Policy learning	-1.40	0.37	-1.37	0.38
FPA	Reward Model	Policy	-1.41	0.27		
VFPA	Variational RM	Variational policy	-1.38	0.39		
V FedBiscuit	V Binary Selector	VRM + VP	-1.36	0.42	-1.28	0.47
VFPA + GP	Variational RM + GB Prior	Variational Policy + GB Prior				

Table 4: Experiments with differential privacy.

Algorithm	FL Training	Centralized Training	10 clients		100 clients	
			Helpfulness $\uparrow$	Harmlessness $\uparrow$	Helpfulness $\uparrow$	Harmlessness $\uparrow$
Original Model			-1.44	0.28		
FedDPO	DPO					
FedBiscuit	Binary Selector	Reward Model + Policy learning	-1.40	0.37	-1.41	0.31
FPA	Reward Model	Policy	-1.41	0.27		
VFPA	Variational RM	Variational policy	-1.38	0.39		
V FedBiscuit	V Binary Selector	VRM + VP	-1.36	0.42	-1.38	0.36
VFPA + GP	Variational RM + GB Prior	Variational Policy + GB Prior				

Table 5: Experiments with differential privacy.

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A APPENDIX

You may include other additional sections here.

Algorithm	FL Training	Centralized Training	Helpfulness $\uparrow$	Honesty $\uparrow$	Truthfulness $\uparrow$
Original Model			-1.25	1.57	2.62
FedBiscuit	Binary Selector	Reward Model + Policy learning	-0.68(+0.57)	1.78(+0.21)	2.82(+0.2)
VFPA	Variational RM	Variational policy	-0.72(+0.53)	1.65(+0.13)	2.77(+0.15)
VFPA + GP	Variational RM + GB Prior	Variational Policy + GB Prior	-0.66(+0.59)	1.82(+0.24)	2.84(+0.22)

Table 6: Experiments with differential privacy.